Robust Estimation of Graph Statistics

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1 Introduction

1.1 Background

1.2 Structure of this paper

1.3 Notations

Definition 1.1. *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community stochastic block model graph with $V = [n]$ and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$. For each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2}) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2}) & \text{if } x_i \neq x_j. \end{cases}$$

Definition 1.2. *The set of positive semidefinite matrices with diagonal entries equal to 1 is denoted by*

$$\mathcal{PSD}_1(n) = \{\boldsymbol{X} \in \mathbb{R}^{n \times n} : \boldsymbol{X} \succeq 0, \boldsymbol{X}_{ii} = 1 \text{ for all } i \in [n]\}.$$

Definition 1.3. *Given a matrix $\boldsymbol{M} \in \mathbb{R}^{n \times n}$, we define $\mathcal{SDP}(\boldsymbol{M}) := \max_{\boldsymbol{X} \in \mathcal{PSD}_1(n)} \langle \boldsymbol{M}, \boldsymbol{X} \rangle$.*

Definition 1.4. *Given a graph G, let $\boldsymbol{A}_G$ be its adjacency matrix, and define $\boldsymbol{A}_G^{\text{cent}} = \boldsymbol{A}_G - \mathbb{E}[\boldsymbol{A}_G]$. For $G \sim \mathcal{G}(n, d, \epsilon)$, we have $\mathbb{E}[\boldsymbol{A}_G] = \frac{d}{n}\mathbf{1}\mathbf{1}^\top$.*

2 (Looking at MS16)

introduce MS16

2.1 Algorithm

Algorithm 1 (SDP Algorithm from [MS16])**.**

1. Sample edges of G independently with probability $(1 + n^{-1/2})^{-1}$ to obtain the sampled edge set $E_{\text{sample}} \subset E$, and then compute the evaluation edge set $E_{\text{eval}} = E \setminus E_{\text{sample}}$.
2. Construct the subgraphs $G_{\text{sample}} = (V, E_{\text{sample}})$ and $G_{\text{eval}} = (V, E_{\text{eval}})$.
3. Compute an optimiser $\boldsymbol{X}^*$ of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$, i.e., $\langle \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle = \mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$.
4. Compute the eigenvectors $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$ of $\boldsymbol{X}^*$.
5. For each $i, j \in [n]$, define $\hat{\boldsymbol{x}}^{(i,j)} \in \{-1, 0, 1\}^n$ as follows:

$$\hat{\boldsymbol{x}}_k^{(i,j)} = \begin{cases} \operatorname{sign}(\boldsymbol{v}_k^{(i)}) & \text{if } |\boldsymbol{v}_k^{(i)}| \geq |\boldsymbol{v}_j^{(i)}|, \\ 0 & \text{otherwise.} \end{cases}$$

6. Find $(i^*, j^*) = \arg\max_{i,j \in [n]} (\hat{\boldsymbol{x}}^{(i,j)})^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)} = \sum_{a,b \in E_{\text{eval}}} \hat{\boldsymbol{x}}_a^{(i,j)} \cdot \hat{\boldsymbol{x}}_b^{(i,j)}$.

7. Output $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G) = \hat{\boldsymbol{x}}^{(i^*,j^*)}$.

We have the following weak recovery rate guarantee for Algorithm 1.

Theorem 2.1 (Weak Recovery for Algorithm 1)**.** *Let $G \sim \mathcal{G}(n, d, \epsilon)$ and assume, for some $\delta > 0$, that $\lambda = \epsilon\sqrt{d}/2 \geq 1 + \delta$. Then, there exists $\eta = \eta(\delta) > 0$ and $d_* = d_*(\delta) > 0$ such that, for all $d \geq d_*$, the output of Algorithm 1, $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G)$, satisfies*

$$\mathbb{P}\left(\frac{1}{n}|\langle \hat{\boldsymbol{x}}^{\mathrm{SDP}}(G), x\rangle| \geq \eta\right) \geq 1 - Ce^{-\sqrt{n}/C}.$$

discuss robustness of the algorithm

3 Proof of Weak Recovery (Theorem 2.1)

Let $\boldsymbol{x}$ be the label vector of $G \sim \mathcal{G}(n, d, \epsilon)$. Note that $G_{\mathrm{sample}} \sim \mathcal{G}(n, d_{\mathrm{sample}}, \epsilon)$, and $G_{\mathrm{eval}} \sim \mathcal{G}(n, d_{\mathrm{eval}}, \epsilon)$, where $d_{\mathrm{sample}} = d/(1 + n^{-1/2})$ and $d_{\mathrm{eval}} = n^{-1/2} d_{\mathrm{sample}}$. For simplicity, from now on we write $d = d_{\mathrm{sample}}$.

introduce the proof strategy

investigate independence

3.1 Step 1: $\boldsymbol{X}^*$ contains nontrivial information about $\boldsymbol{x}$

We first introduce a group of Gaussian random matrices (GOEs) that have well-behaved SDP values.

Definition 3.1 (Gaussian orthogonal ensemble (GOE) and the deformed GOE $\boldsymbol{B}(\lambda)$)**.** *A random matrix $\boldsymbol{W} \in \mathbb{R}^{n\times n}$ is a* GOE matrix *if it is a symmetric matrix with $\{\boldsymbol{W}_{ij}\}_{i\leq j}$ independent and $\boldsymbol{W}_{ij} \sim \mathcal{N}(0, 1/n)$ for $i < j$ and $\boldsymbol{W}_{ii} \sim \mathcal{N}(0, 2/n)$. The* deformed GOE matrix $\boldsymbol{B}(\lambda)$ *is then defined as*

$$\boldsymbol{B}(\lambda) = \frac{\lambda}{n}\mathbf{1}\mathbf{1}^\top + \boldsymbol{W}.$$

We then have the following phase transition result for the SDP value of $\boldsymbol{B}(\lambda)$.

Lemma 3.1 (SDP value of $\boldsymbol{B}(\lambda)$, Theorem 5 of [MS16])**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix defined in Definition 3.1. Then, with probability converging to 1 as $n \to \infty$,*

(a) if $\lambda \leq 1$: for any $\epsilon > 0$, we have $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \in [2 - \epsilon, 2 + \epsilon]$.

(b) if $\lambda > 1$: there exists $\Delta(\lambda) > 0$ such that $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \geq 2 + \Delta(\lambda)$.

If we set the diagonal entries of $\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{sample}}}$ to be $\lambda\sqrt{d}/n$, then

$$\frac{1}{\sqrt{d}}\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{sample}}} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{E},$$

where $\boldsymbol{E}$ is a symmetric matrix with zero diagonal and independent off-diagonal entries satisfying

$$\boldsymbol{E}_{ij} = \begin{cases} \frac{1}{\sqrt{d}} - \frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } \mathbb{P}((i,j) \in E_{\mathrm{sample}}), \\ -\frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } 1 - \mathbb{P}((i,j) \in E_{\mathrm{sample}}). \end{cases}$$

Let $\boldsymbol{X}^*$ be an optimizer of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$. We will now try to bound $\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle$ and $\langle \frac{1}{\sqrt{d}} \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle$ away from each other using the phase transition result of Lemma 3.1, which will then give us a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$, which is the correlation between $\boldsymbol{X}^*$ and the ground truth labels matrix $\boldsymbol{x}\boldsymbol{x}^\top$.

The following lemma bounds $\mathcal{SDP}(\boldsymbol{E})$ to the SDP value of a GOE matrix.

Lemma 3.2 (GOE approximation of $\mathcal{SDP}(\boldsymbol{E})$)**.** *Let $\boldsymbol{W}$ be a GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$ with probability at least $1 - Ce^{-n/C}$, we have*

$$\left| \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We also have a similar result bounding $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$ by the SDP value of a deformed GOE matrix.

Lemma 3.3 (Deformed GOE approximation of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$)**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$*

$$\left| \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We will discuss the proofs of the above two lemmas later in Section 4.1.1. We will take them for granted here.

Now we have the tools to obtain a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$.

$$\begin{aligned}
\left\langle \frac{\lambda}{n^2}\boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \right\rangle &= \frac{1}{n}\left(\left\langle \frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \right\rangle - \langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \right) \\
&= \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \\
&\geq \frac{1}{n}\mathcal{SDP}(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) && \text{by optimality of SDP} \\
&\geq \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) - \frac{C\log(d)}{d^{1/10}} && \text{by Lemmas 3.2 and 3.3} \\
&\geq \Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}}, && \text{by Lemma 3.1}
\end{aligned}$$

with probability at least $1 - Cn^{-4}$ (see the remark below).

Remark 3.1. In the original paper [MS16], the authors claimed that the above inequality holds with probability at least $1 - Ce^{-n/C}$, but tracking the proof of Lemma 3.1 (Appendices F and G of [MS16]) shows that they only proved the result with probability at least $1 - Cn^{-4}$ (Lemma G.8 of [MS16]). This is still sufficient for our purposes, as the probability still tends to 1 as $n \to \infty$.

Now fix λ, and take d sufficiently large such that $\Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}} \geq \lambda\Delta_2(\lambda) > 0$. Then, we have, with

probability at least $1 - Cn^{-4}$,

$$n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle \geq \Delta_2(\lambda).$$

This is an important milestone, as it shows that $\boldsymbol{X}^*$ has a nontrivial correlation with the ground truth $\boldsymbol{x}\boldsymbol{x}^\top$.

3.2 Step 2: There exists "good" label estimators among $\{\hat{\boldsymbol{x}}^{(i,j)}\}_{i,j\in[n]}$

We first define the following rounding function used in step 5 of Algorithm 1:

Definition 3.2 (Rounding function)**.** *For $t \geq 0$, define the* rounding function $r_t : \mathbb{R} \to \{-1, 0, +1\}$ *as follows:*

$$r_t(x) = \begin{cases} 1 & \text{if } x \geq t, \\ 0 & \text{if } -t < x < t, \\ -1 & \text{if } x \leq -t. \end{cases}$$

Hence, we can write $\hat{\boldsymbol{x}}^{(\hat{\imath},j)} = r_{\boldsymbol{v}_j^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})})$.

Since $\boldsymbol{X} \in \mathcal{PSD}_1(n)$, writing its eigen-decomposition as $\boldsymbol{X} = \sum_{i=1}^n \xi_i \boldsymbol{v}^{(i)}(\boldsymbol{v}^{(i)})^\top$, where ξ_i is the i-th largest eigenvalue, and $\boldsymbol{v}^{(i)}$ is the corresponding unit length eigenvector, we have $\xi_i \geq 0$ for all i by positive semidefiniteness, and $\sum_{i=1}^n \xi_i = \mathrm{Tr}(\boldsymbol{X}) = n$. Thus,

$$\Delta_2(\lambda) \leq \frac{1}{n^2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle = \frac{1}{n^2}\sum_{i=1}^n \xi_i(\boldsymbol{x}^\top \boldsymbol{v}^{(i)})^2 = \sum_{i=1}^n \frac{\xi_i}{n} \cdot \left(\frac{\boldsymbol{x}^\top \boldsymbol{v}^{(i)}}{\sqrt{n}}\right)^2.$$

So there exists $\hat{\imath} \in [n]$ such that $|\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})}| \geq \sqrt{n\Delta_2(\lambda)}$.

Next, we will use following lemma to justify that the rounding function preserves this correlation, which shows that it is possible to extract a good label estimator.

Lemma 3.4 (Rounding Lemma)**.** *Let X, Y be random variables such that: $\mathbb{P}(X = +1) = \mathbb{P}(X = -1) = 1/2$, $\mathbb{E}(XY) \geq \epsilon > 0$, and $\mathbb{E}(Y^2) = 1$. Then, there exists t^* depending on the joint distribution of X and Y, such that*

$$\mathbb{E}(Xr_{t^*}(Y)) \geq \frac{\epsilon^2}{2}.$$

Proof. Note that for all t, we have $Xr_t(Y) = r_t(XY)$, and $Y^2 = (XY)^2$. Let $Z = XY$, then it suffices to show that given $\mathbb{E}(Z) \geq \epsilon > 0$ and $\mathbb{E}(Z^2) = 1$, we have $\mathbb{E}(r_t(Z)) \geq \epsilon^2/2$. Recall that for nonnegative random variable S, by Fubini, we have

$$\mathbb{E}(S) = \mathbb{E}\left(\int_0^\infty \mathbb{1}_{s\leq S} ds\right) = \int_0^\infty \mathbb{P}(S \geq s)ds, \quad \mathbb{E}(S^2) = \mathbb{E}\left(\int_0^\infty 2s\mathbb{1}_{s\leq S} ds\right) = \int_0^\infty 2s\mathbb{P}(S \geq s)ds.$$

Therefore, writing $Z^{\pm} = \max(\pm Z, 0)$, we have

$$\begin{aligned}
\epsilon \le \mathbb{E}(Z) = \mathbb{E}(Z_+) - \mathbb{E}(Z_-) &= \int_0^\infty \mathbb{P}(Z_+ \ge z)dz - \int_0^\infty \mathbb{P}(Z_- \ge z)dz \\
&\le \int_0^K (\mathbb{P}(Z \ge z) - \mathbb{P}(Z \le -z))dz + \int_K^\infty \frac{2z}{2K}\mathbb{P}(Z \ge z)dz - \int_K^\infty \frac{2z}{2K}\mathbb{P}(-Z \ge z)dz \\
&= \int_0^K \mathbb{E}r_t(Z)dt + \frac{1}{2K}\mathbb{E}(Z_+^2 - Z_-^2) = \int_0^K \mathbb{E}r_t(Z)dt + \frac{1}{2K}.
\end{aligned}$$

Now take $K = 1/\epsilon$, then

$$\frac{1}{K}\int_0^K \mathbb{E}r_t(Z)dt \ge \frac{1}{K}(\epsilon - \frac{1}{2K}) = \frac{\epsilon^2}{2}$$

So the average of $\mathbb{E}r_t(Z)$ over $t \in [0, 1/\epsilon]$ is greater than $\epsilon^2/2$, so there exists $t^* \in [0, 1/\epsilon]$ such that $\mathbb{E}r_{t^*}(Z) \ge \epsilon^2/2$. □

Without loss of generality, assume that $\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} > 0$. Consider (X, Y) drawn uniformly from $\{(\boldsymbol{x}_k, \sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})})\}_{k\in[n]}$, then we have $\mathbb{E}(XY) = n^{-1/2}\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} \ge \sqrt{\Delta_2(\lambda)}$ and $\mathbb{E}(Y^2) = n^{-1}||\sqrt{n}\boldsymbol{v}^{(\hat{\imath})}||^2 = ||\boldsymbol{v}^{(\hat{\imath})}||^2 = 1$. Thus, by Lemma 3.4, there exists $t^* \in [0, 1/\sqrt{\Delta_2(\lambda)}]$ such that

$$\frac{\Delta_2(\lambda)}{2} \le \mathbb{E}(Xr_{t^*}(Y)) = \frac{1}{n}\sum_{k=1}^n \boldsymbol{x}_k r_{t^*}(\sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})}) = \frac{1}{n}\langle \boldsymbol{x}, r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})})\rangle,$$

where $\tilde{t} = t^*/\sqrt{n} \in [0, 1/\sqrt{n\Delta_2(\lambda)}]$.

Remark 3.2. Here, we slightly improved the rounding lemma bound from $\epsilon^2/4$ in [MS16] to $\epsilon^2/2$, although the main proof idea remains the same. Also, the original application of the rounding lemma missed the scaling factor $\sqrt{n}$ in the definition of Y, but the conclusion remains valid.

Finally, note that for any value of $\tilde{t}$, we can find $\hat{\jmath} \in [n]$ such that $r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})}) = r_{\boldsymbol{v}_{\hat{\jmath}}^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})}) = \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}$. Therefore, there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle \ge \frac{\Delta_2(\lambda)}{2}. \tag{3.1}$$

Now, it remains to show that the selection step (Step 6) of Algorithm 1 will find such $\hat{\imath}, \hat{\jmath}$ with high probability, which will then complete the proof of Theorem 2.1.

3.3 Step 3: A "good" estimator can be found with G_{eval}

We first want to relate the quantities $\hat{\boldsymbol{x}}^\top \boldsymbol{A}_{G_{\text{eval}}}^{\text{cent}}\hat{\boldsymbol{x}}$, the objective function in Step 6 of Algorithm 1, to $\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle$, the correlation above, with high probability. Recall the well-known Azuma-Hoeffding inequality.

add reference

Lemma 3.5 (Azuma-Hoeffding inequality)**.** *Let $X_0, \ldots, X_m$ be a martingale, with bounded increments, i.e.*

$|X_i - X_{i-1}| \le c_i$ *for all* $i \in [m]$*. Then, for all* $t > 0$*, we have*

$$\mathbb{P}(|X_m - \mathbb{E}(X_m)| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m c_i^2}\right).$$

We apply this inequality to the objective function in Step 6 of Algorithm 1, and obtain the following concentration result.

Lemma 3.6. *Let* $G \sim \mathcal{G}(n, d, \epsilon)$*. For any fixed vector* $\hat{\boldsymbol{x}} \in \{-1, 0, +1\}^n$*, define* $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}}$*. Then, for any* $t > 0$*,*

$$\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E_{\text{eval}}| \le m\Big) \le 2\exp\left(-\frac{t^2}{8m}\right).$$

Proof. Note that $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}} - (d/n)(\hat{\boldsymbol{x}}^\top \mathbf{1})^2$, so writing $\tilde{f}(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}}$, we have $f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}})) = \tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))$. Note that $\tilde{f}(G; \hat{\boldsymbol{x}}) = \sum_{(a,b) \in E} 2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b$.

For some ordering of all possible edges, we can define a filtration $\mathcal{F}_0, \mathcal{F}_1, \ldots, \mathcal{F}_m$ with $\mathcal{F}_k = \sigma(E_1, \ldots, E_k)$, where $E_k \in E(G) \cup \varnothing$ is the k-th edge of G in the ordering. Then, define the martingale $X_k = \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | \mathcal{F}_k)$ for $k \in [m]$. Since $|E(G)| \le m$, we have $X_m = \tilde{f}(G; \hat{\boldsymbol{x}})$. Note that for all $k \in [m]$, we have

$$\begin{aligned}
|X_k - X_{k-1}| &= \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1})\Big| \\
&\le \sup_{e_{1:k-1}, e_k, e_k'} \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k')\Big| \\
&\le \sup_{E(G^+) = E(G) \cup \{(a,b)\}} \Big|\tilde{f}(G^+; \hat{\boldsymbol{x}}) - \tilde{f}(G; \hat{\boldsymbol{x}})\Big| \\
&\le \sup_{a,b} |2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b| \le 2.
\end{aligned}$$

Hence, we can apply the Azuma-Hoeffding inequality Lemma 3.5 to X with $c_i = 2$ for all $i \in [m]$ to obtain

$$\begin{aligned}
\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E(G)| \le m\Big) &= \mathbb{P}\Big(\big|\tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E(G)| \le m\Big) \\
&= \mathbb{P}(|X_m - \mathbb{E}(X_m)| \ge t) \\
&\le 2\exp\left(-\frac{t^2}{8m}\right).
\end{aligned}$$

as desired. □

To apply the above concentration result, we need to bound $|E_{\text{eval}}| \sim \text{Binomial}\left(|E|, n^{-1/2}/(1 + n^{-1/2})\right)$, which we can achieve with the following Chernoff bound argument.

Lemma 3.7 (Bounding $|E_{\text{eval}}|$)**.** *Let* $p_n = n^{-1/2}/(1 + n^{-1/2})$*. Then, for some constant* $C > 0$*, we have*

$$\mathbb{P}(|E_{\text{eval}}| \ge C\sqrt{n}) \le Ce^{-\sqrt{n}/C}.$$

Proof. Recall that $|E| \sim \mathrm{Binomial}\left(\binom{n}{2}, d/n\right)$, so by Chernoff bound, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E| \geq K_0 n) &\leq e^{-sK_0 n}\mathbb{E}(e^{s|E|}) = e^{-sK_0 n}\left(1 - \frac{d}{n} + \frac{d}{n}e^s\right)^{\binom{n}{2}} \\
&\leq \exp\left(-sK_0 n + \frac{d}{n}(e^s - 1)\binom{n}{2}\right) \\
&\leq \exp\left(\left(\frac{d}{2}(e^s - 1) - sK_0\right)n\right) \\
&\leq C_0 e^{-n/C_0},
\end{aligned}$$

where the last inequality holds when K_0 is sufficiently large, and C_0 is a constant depending on d.

Now, on the event $\{|E| \leq K_0 n\}$ with probability at least $1 - C_0 e^{-n/C_0}$, we have $|E_{\mathrm{eval}}| \sim \mathrm{Binomial}\left(|E|, p_n\right)$, so by Chernoff bound again, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E_{\mathrm{eval}}| \geq K_1\sqrt{n}) &= \mathbb{E}\left(\mathbb{P}(|E_{\mathrm{eval}}| \geq K_1\sqrt{n} \mid |E|)\right) \leq e^{-sK_1\sqrt{n}}\mathbb{E}(\mathbb{E}(e^{s|E_{\mathrm{eval}}|} \mid |E|)) \\
&= e^{-sK_1\sqrt{n}}\mathbb{E}((1 - p_n + p_n e^s)^{|E|}) \leq e^{-sK_1\sqrt{n}}\left(1 + p_n(e^s - 1)\right)^{K_0 n} \\
&= \exp\left(-sK_1\sqrt{n} + K_0(e^s - 1)np_n\right) \leq \exp\left((K_0(e^s - 1) - sK_1)\sqrt{n}\right) \\
&\leq C_1 e^{-\sqrt{n}/C_1},
\end{aligned}$$

where the last inequality holds when K_1 is sufficiently large, and C_1 is a constant depending on d. Combining the two probability bounds, we have

$$\mathbb{P}(|E_{\mathrm{eval}}| \geq K\sqrt{n}) \leq C_0 e^{-n/C_0} + C_1 e^{-\sqrt{n}/C_1} \leq C e^{-\sqrt{n}/C},$$

for suitable constants C, as desired. □

expand the following argument slightly

By Lemmas 3.6 and 3.7, for all $s \in [0, 1]$, we have, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$,

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}} - \mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}}\right)\right| \leq s\sqrt{n}.$$

Note that

$$\begin{aligned}
\mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}}\right) &= \hat{\boldsymbol{x}}^\top \mathbb{E}\left(\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\right)\hat{\boldsymbol{x}} \\
&= \hat{\boldsymbol{x}}^\top \frac{\epsilon d}{2n}\left(\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{I}_n\right)\hat{\boldsymbol{x}} \\
&= \frac{\epsilon d}{2n}(\hat{\boldsymbol{x}}^\top \boldsymbol{x})^2 - \frac{\epsilon d}{2}.
\end{aligned}$$

Thus, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$, we have

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}} - \frac{\epsilon d}{2n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle^2 + \frac{\epsilon d}{2}\right| \leq s\sqrt{n}. \tag{3.2}$$

This concentration result relates $\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}$ to the correlation $\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle$ with high probability.

In particular, at the end of the previous step (Equation (3.1)), we have shown that there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \rangle \geq \frac{\Delta_2(\lambda)}{2}$. Then, by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$ (taking $s = 1$), we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \rangle\right)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{8}\Delta_2(\lambda)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{12}\Delta_2(\lambda)^2 =: \Delta_3(\lambda),$$

where the last inequality follows from

$$\frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{24}\Delta_2(\lambda)^2,$$

which holds for sufficiently large n. Therefore,

$$\max_{i,j\in[n]} \frac{1}{n}\hat{\boldsymbol{x}}^{(i,j)\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)} \geq \frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \Delta_3(\lambda).$$

On the other hand, suppose that $\hat{\boldsymbol{x}}$ has a lower correlation with $\boldsymbol{x}$, say $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle \leq \frac{\Delta_2(\lambda)}{4}$, then by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$, we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}} \leq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle\right)^2 + \frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{32}\Delta_2(\lambda)^2 + \frac{\epsilon d}{24}\Delta_2(\lambda)^2 < \frac{\epsilon d}{12}\Delta_2(\lambda)^2 = \Delta_3(\lambda).$$

Thus, with probability at least $1 - Ce^{-\sqrt{n}/C}$, the maximizer $\hat{\boldsymbol{x}}^{\text{SDP}} = \hat{\boldsymbol{x}}^{(i^*,j^*)}$ of $\max_{i,j\in[n]} \hat{\boldsymbol{x}}^{(i,j)\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)}$ must satisfy

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{\text{SDP}} \rangle \geq \frac{\Delta_2(\lambda)}{4}.$$

This completes the proof of Theorem 2.1. ■

4 Examining Robustness Properties of the SDP

4.1 Approximating SDP of adjacency matrices / Looking at the unproved lemmas and extension to graphons

We will now briefly sketch out the proofs of Lemmas 3.2 and 3.3 by highlighting the main ideas (see the original paper [MS16] for details). We will then apply these ideas to show that the algorithm works for a more general class of random graph models, which we will define below.

Definition 4.1 (Perturbed SBM)**.** *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community h-perturbed stochastic block model graph with $V = [n]$, uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$, i.i.d. latent variables $\xi_i \sim \text{Unif}([0,1])$ for $i \in [n]$, and perturbation function $h : [0,1]^2 \to \mathbb{R}$ satisfying*

$\mathbb{E}[h(\xi_i, \xi_j)] = 0$ *for all* $i, j \in [n]$*, if for each pair of vertices* $i, j \in [n]$*, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2} + h(\xi_i, \xi_j)) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2} + h(\xi_i, \xi_j)) & \text{if } x_i \neq x_j. \end{cases}$$

Remark 4.1. Note that the proof of weak recovery in Section 3 uses the adjacency matrix of the graph only in two places: the SDP approximation lemmas (Lemmas 3.2 and 3.3), which we will discuss for the rest of this section, and $\mathbb{E}[\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}]$ in Equation (3.2), which will be the same for the perturbed SBM and the original SBM since $\mathbb{E}[h(\xi_i, \xi_j)] = 0$ for all $i, j \in [n]$. Thus, to show that the algorithm still works for the perturbed SBM, we only need to show that similar SDP approximation results still hold for the perturbed SBM.

4.1.1 General approach to approximating SDP of adjacency matrices

The main idea of approximating the SDP of adjacency matrices is to utilize the following two helper functions:

Definition 4.2 ($\mathcal{OPT}_k$)**.** *Define the rank-constrained SDP value for* $\boldsymbol{M} \in \mathbb{R}^{n \times n}$ *and* $k \in [n]$*:*

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\substack{\boldsymbol{X} \in \mathcal{PSD}_1(n) \\ \text{rank}(\boldsymbol{X}) \leq k}} \langle \boldsymbol{M}, \boldsymbol{X} \rangle.$$

Equivalently, let $\mathbb{S}^{k-1} = \{\boldsymbol{v} \in \mathbb{R}^k : ||\boldsymbol{v}||_2 = 1\}$ *be the unit sphere in* $\mathbb{R}^k$*, then we have*

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_n \in \mathbb{S}^{k-1}} \sum_{i,j=1}^{n} M_{ij} \langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j \rangle.$$

Definition 4.3 (Smoothed OPT)**.** *For* $\beta > 0$*, define* $\Phi(\beta, k; \boldsymbol{M})$*, the smoothed version of* $\mathcal{OPT}_k(\boldsymbol{M})$*, as follows:*

$$\Phi(\beta, k; \boldsymbol{M}) = \frac{1}{\beta} \log \left(\int_{(\mathbb{S}^{k-1})^n} \exp \left(\beta \sum_{i,j=1}^{n} M_{ij} \langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j \rangle \right) d\nu(\boldsymbol{\sigma}) \right),$$

where ν *is the uniform probability measure on* $(\mathbb{S}^{k-1})^n$*.*

We can see that as $\beta \to \infty$*,* $\Phi(\beta, k; \boldsymbol{M})$ *will select the maximum value of* $\sum_{i,j=1}^{n} M_{ij} \langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j \rangle$*, which is exactly* $\mathcal{OPT}_k(\boldsymbol{M})$*.*

In their paper, Montanari and Sen showed the following bounds relating $\mathcal{SDP}(\boldsymbol{M})$, $\mathcal{OPT}_k(\boldsymbol{M})$ and $\Phi(\beta, k; \boldsymbol{M})$. We will quote these results without proof.

Theorem 4.1 (SDP vs OPT, Theorem 4 Equation 15 of [MS16])**.** *For* $k \geq k_0$*, where* k_0 *is a sufficiently large constant, for any symmetric matrix* $\boldsymbol{M} \in \mathbb{R}^{n \times n}$*, we have*

$$\left(1 - \frac{1}{k}\right) \mathcal{SDP}(\boldsymbol{M}) - \frac{1}{k} |\mathcal{SDP}(-\boldsymbol{M})| \leq \mathcal{OPT}_k(\boldsymbol{M}) \leq \mathcal{SDP}(\boldsymbol{M}).$$

Theorem 4.2 (OPT vs Φ, Lemma 3.2 of [MS16]). *If $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ satisfies*

$$\|\boldsymbol{M}\|_{\infty\to 2} := \max_{||\boldsymbol{v}||_\infty \le 1} ||\boldsymbol{M}\boldsymbol{v}||_2 \le L\sqrt{n},$$

then for any $\beta > 0$, $k \in [n]$, and $\epsilon \in (0,1]$, there exists a constant C, such that

$$\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \frac{1}{n}\mathcal{OPT}_k(\boldsymbol{M})\right| \le 2L\epsilon\sqrt{k} + \frac{k}{\beta}\log\frac{C}{\epsilon}.$$

briefly bound SDP to Phi for B(lambda) and cent adj matrix

4.1.2 Interpolation

References

[MS16] Montanari, A. and Sen, S. (2015) *Semidefinite Programs on Sparse Random Graphs and their Application to Community Detection*, Proceedings of the forty-eighth annual ACM symposium on Theory of Computing, 2016, pp. 814–827.

[1] Author (Year) *Title*, Publisher